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Noise by Choice: AR-Mediated Selective Noise Control for Office Environment

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Noise is a major annoyance in our everyday life, where active noise canceling (ANC) headphones is a common solution to remove noise. However, it also removes all the sounds in the environment. Particularly the widely adopted open plan offices, noise is a primary source that contributes to work dissatisfaction in offices, which the use of ANC headphones forces workers to choose between suffering the noise and isolating themselves. Augmented reality (AR) mediated selective noise control that allow users to selectively reduce or mute annoying sound while preserving valuable ones presents a middle ground. To implement AR-mediated selective noise control systems, we need to understand how such a system should be designed and how it fits into a real-world context, specifically the workplace, through a human-centered design approach. Two interaction principles for selective noise control interface design are grounded in the ”where” and ”what” auditory processing streams, where the spatial principle is based on selecting sounds by their location, and the semantic principle is based on selecting sounds by predefined categories. We conducted a mixed-methods within-subjects study with 20 participants comparing spatial and semantic interaction principles for selective noise control, and understanding how they perceive using such a system in their real office context. Interfaces of the two interaction principles are tested in a VR-simulated office to evaluate their usability and cognitive load measures. An interview is conducted after the testing to understand what factors of such a system influence these design measures and how they perceive such a system being used in their workplace. Both principles carry distinct trade-offs. The spatial interface achieved higher usability and lower cognitive load due to its intuitive object-based interaction. Meanwhile, the semantic interface was preferred when real-world use was considered, where efficiency and social acceptability were prioritized. Participants found selective noise control beneficial as it offers the agency and social awareness that ANC headphones cannot provide, while raising concerns around adjustment workload, social dynamics, and privacy. We contribute interface and system design suggestions alongside social and ethical considerations for future development and deployment of selective noise control systems in office environments.

SAMMANFATTNING

Buller är ett stort irritationsmoment i vår vardag, där hörlurar med aktiv brusreducering (ANC) är en vanlig lösning för att eliminera oönskat ljud men det tar…Buller är ett stort irritationsmoment i vår vardag, där hörlurar med aktiv brusreducering (ANC) är en vanlig lösning för att eliminera oönskat ljud men det tar även bort alla ljud från omgivningen. Särskilt i de allt vanligare kontorslandskapen är buller en av de främsta källorna till arbetsmissnöje, vilket tvingar arbetstagare att välja mellan att utstå bullret och att isolera sig från sin omgivning. Förstärkt verklighet (AR) som möjliggör selektiv ljudkontroll, där användare kan dämpa eller tysta störande ljud samtidigt som värdefulla ljud bevaras, utgör en mellanväg. För att implementera AR-baserade system för selektiv ljudkontroll behöver vi förstå hur ett sådant

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system bör utformas och hur det passar in i ett verkligt sammanhang, specifikt på arbetsplatsen, genom en människocentrerad designprocess. Två interaktionsprinciper för gränssnitt för selektiv ljudkontroll grundar sig i de auditiva bearbetningsströmmarna ”var” och ”vad”, där den spatiala principen bygger på att välja ljud utifrån deras plats, och den semantiska principen bygger på att välja ljud utifrån fördefinierade kategorier. Vi genomförde en mixad-metod-studie med upprepade mätningar med 20 deltagare, där vi jämförde spatiala och semantiska interaktionsprinciper för selektiv ljudkontroll och undersökte hur deltagarna uppfattar användningen av ett sådant system i sitt verkliga kontorssammanhang. Gränssnitten för de två interaktionsprinciperna testades i ett VR-simulerat kontor för att utvärdera deras användbarhet och kognitiv belastning. Efter testningen genomfördes en intervju för att förstå vilka faktorer i ett sådant system som påverkar dessa designmått och hur deltagarna ser på användningen av systemet på sin arbetsplats. Båda principerna har tydliga avvägningar: det spatiala gränssnittet uppnådde högre användbarhet och lägre kognitiv belastning tack vare sin intuitiva objektbaserade interaktion, medan det semantiska gränssnittet föredrogs när verklig användning beaktades, där effektivitet och social acceptabilitet prioriterades. Deltagarna upplevde selektiv ljudkontroll som fördelaktigt eftersom det ger handlingsutrymme och social medvetenhet som ANC-hörlurar inte kan erbjuda, samtidigt som de lyfte farhågor kring arbetsbelastning vid justering, sociala dynamiker och integritet. Vi bidrar med förslag på gränssnitts- och systemdesign samt sociala och etiska överväganden för framtida utveckling och driftsättning av system för selektiv ljudkontroll i kontorsmiljöer.

CCS Concepts: • **Human-centered computing** → **Ubiquitous and mobile computing**; **Interaction design**; *Mixed / augmented reality*; • **Applied computing** → Sound and music computing.

Keywords: audio augmented reality, selective noise control, open-plan office, spatial interaction, semantic interaction, workplace acoustics

Nyckelord: ljudaugmenterad verklighet, selektiv bullerkontroll, öppet kontorslandskap, spatial interaktion, semantisk interaktion, arbetsplatsakustik

1 INTRODUCTION

With the current development of Augmented Reality and auditory technology, selective noise control that allows adjustments of individual sounds in one’s environment, rather than blocking all noise, is becoming increasingly possible to achieve. This presents an opportunity for improving acoustic experiences by allowing individuals to hear desired sounds while removing undesired ones. Noise is a major source of annoyance, where in work place, it is one of the primary sources of work dissatisfaction [40]. The dominant open plan office design contributes particularly to poor acoustic environments due to its minimal physical barriers [20]. Yet, current solutions force workers into a trade-off, where they either need to endure the noise or isolate themselves with Active Noise Canceling (ANC) headphones that sacrifice their social awareness, which makes shared workspaces valuable [81]. Therefore, AR-mediated selective noise control offers a middle ground as it allows users to manage their sonic environment without complete auditory isolation. However, designing such systems requires understanding how users should interact with sound sources, and how such system would fit into their daily work life.

AR-mediated selective noise control can use visual interfaces for the selection and adjustment of sound sources. The interface can be designed according to the two distinct processing streams of how humans process sounds, either the ”where” stream for localizing sounds spatially or the ”what” stream

for identifying their meaning semantically [74]. However, each process leads to inherent design characteristics and constraints, leading to two distinct interaction design principle: spatial and semantic principles. Existing systems like Aurchestra [65] and Semantic Hearing [90] have been designed with 2D screen-based semantic interfaces, and other systems adopt 3D spatial interfaces, such as MoXaRt [97] and AudioMiXR [93]. However, as each focuses on a single interface design, how the two principles differ in user experience remains unexplored.

Beyond interaction design, selective noise control can have broader implications on workers' experience when used in their workplace. Unlike ANC headphones that offer only binary control and create auditory isolation [81], selective noise control has the potential to preserve social presence and give workers agency over their soundscape while addressing noise discomfort. However, how users perceive the role of such a system in their real workplace, including its potential benefits, how it fits into their workflow, and possible concerns, remains an open question.

In order to investigate the interface design of the selective noise control system and how they are perceived when used in a real office context, this paper addresses the following research questions:

> RQ1: How do the spatial and semantic interaction principles differ in user experience for selective noise control?
>
> RQ2: How do users their usage of selective noise control system in their real work context?

To address these questions, a mixed-methods study was conducted with 20 participants in a VR simulated open plan office environment, where participants adjusted the volume of simulated office sounds using two interfaces based on each interaction principle, followed by an interview. RQ1 was addressed through both usability and cognitive load scoring and the first part of the interview. Results showed that the spatial interface achieved higher usability and lower cognitive load due to its intuitive object-based interaction, while the semantic interface was preferred when real-world use was considered, as efficiency and social acceptability were prioritized. RQ2 was addressed through the second part of the interview. Findings revealed participants perceived selective noise control as beneficial over ANC headphones for preserving social awareness and offering greater agency, while raising concerns around adjustment workload, social dynamics, privacy, and potential workplace harassment. This work contributes (1) an empirical comparison of spatial and semantic interaction principles for selective noise control, (2) insights into how users perceive such systems in a real workplace context, and (3) design, social, and ethical considerations for future selective noise control systems deployed in office environments.

2 RELATED WORK

2.1 Office Noise and Current Solutions

In recent years, open-plan offices have been the major office design trend as it is intended to improve communication and collaboration by allowing more visual contact between coworkers [20, 36, 40]. However, empirical evidence suggests that such office design is not beneficial to workers due to how it decreases concentration and productivity, increases stress, reduces privacy, and lowers overall job satisfaction, with the high level of noise being the key contributing factor [20, 36, 38, 40, 70, 71]. This high noise level is caused by the minimal physical barriers used for open-plan offices' design, which includes interior walls that allows physical sound insulation are removed [20, 36, 38].

Research in office acoustic environment have identified major types of noise sources, which includes speech, notifications (e.g. phone ring, message notifications), sounds from office machinery and electronic (e.g. ventilation systems, printer, computer), outdoor sounds, and other human made sounds (e.g.

footsteps, laughter, rolling chairs) [1, 17, 20, 42, 70, 71]. Irrelevant and intelligible speech is considered the most annoying noise, including discussion between multiple people and call conversations with one person's voice, as an individual can understand the meaning of the speech and unconsciously process that information, leading to high cognitive disturbance [7]. Meanwhile, constant ambient noises, such as ventilation and outdoor sounds, are considered to be less annoying, and sometimes even seen as favorable, as individuals can get used to such ambient noises after prolonged exposure, and it can avoid complete silence that leads to more discomfort [1, 43]. Other than the specific types of noise, how well one can predict and control the sound also influences the annoyance level, as when the person is creating the noise themselves, they are less annoyed by it [43], suggesting that giving workers agency over their soundscape may itself reduce annoyance.

There are two major approaches for reducing the noise level in offices. One is installing physical sound absorbing materials, such as false ceiling and dividing panels [17]. However, these sound-absorbing materials mainly reduce outdoor noise that is relatively low volume, but not efficient for the higher volume noises happening inside the office [17, 20, 57, 70, 71]. Another approach is wearing active noise-cancelling (ANC) headphones. Studies have shown that the use of ANC headphones with masking sound, such as music and white noise, can allow better focus, reduce stress, and improve an individual's work satisfaction [57, 73]. However, using ANC headphones can create an isolated auditory environment, causing workers to detach from the environment [57, 81]. This isolation, other than losing awareness to important auditory cues (e.g. alarms, warnings), it also creates social disconnection [83]. Such disconnection can lead to a loss of social interactions, such as collaboration and small talk, potentially impacting coworker relationships [81]. AR-mediated selective noise control presents a potential alternative to ANC headphones that addresses both the noise discomfort and the social isolation. However, designing such a system requires understanding how users should interact with sound sources and whether different interaction design principles differ in user experience. Beyond interface design, how users perceive such a system fitting into their real workplace context, including its potential benefits, drawbacks, and concerns, also remains as open questions that is addressed in this work.

2.2 Audio Augmented Reality

Augmented Reality (AR) extends beyond visual augmentation to include all sensory modalities, including auditory perception [13, 51, 54, 87, 98]. While most AR research focuses on augmenting visual elements onto real-world environments [30, 76, 86], it also considers the modification of real-world sensory perception, including the reduction or removal of real-world elements (i.e. Diminished Reality) [3, 55, 68, 76]. Selective noise control falls under auditory diminished reality, where sounds in the real environment are reduced or removed rather than augmented [91]. Audio Augmented Reality (AAR) is the broader term that contains both the augmentation and modification of people's real sound environment [13, 28, 56, 98].

AAR can be achieved by using hearables (i.e. personal hearing devices) or surrounding speakers [27, 62, 98]. Hearables are suitable for personal AAR systems that require acoustic hear-through capabilities that allow users to hear the already existing sounds in the real environment while adding virtual sounds and/or modifying the existing sounds [4, 98]. These hearables can be divided into two categories: (1) occluding devices that block sound waves from entering the ear but have acoustic transparency capability to allow real sounds to pass through, and (2) floating-type devices that do not touch the ear directly, where the real sounds are not completely blocked [27, 98]. Although hear-through systems have allowed more

AAR capability, they have historically been constrained by processing latency, as real sounds must be captured, processed, and replayed in real time, limiting the feasibility of early personal AAR systems [27, 35, 98].

Bhattacharyya et al. [4] distinguish two types of auditory linkage in AAR. The first linkage is *sonic linking*, where specific real-world sound events serve as input for soundscape modification, and the second is acoustic linking, where the overall acoustic environment is treated as input. Selective noise control falls under the sonic linking paradigm, as it targets individual sound sources for adjustment rather than modifying the soundscape as a whole. Recent AAR research has also explored the types of augmentation and modification possible with AAR [10, 13, 44, 98], as well as users' soundscape customization preferences in everyday life [10, 28, 86] and in specific settings, such as home [33, 37, 67] and workplace [15]. Across these works, selective noise adjustment emerges as a recurring user desire [10, 28, 33, 86]. Specifically, Bustoni et al. [10] have investigated user preference of different selective noise altering and masking strategy, including volume reduction and complete removal of noise, for different sound categories.

AAR technology has matured significantly in recent years. Machine learning models can now separate sounds by their spatial location or semantic classification [16, 50, 60, 88, 96], with Xiao et al.'s [95] work on spatially selective active noise canceling demonstrating that targeted sound source adjustment is technically realizable. Combined with continued advancements in hearable technology with better acoustic hear-through capabilities, lower latency, and higher audio fidelity, there is a clear technological feasibility of AR-mediated selective noise control, motivating the investigation of how such systems should be designed and experienced by users. This work investigates a system that can be achieve in the near future with this technological feasibility by simulating such a system in VR to understand how it should be designed and experienced, without being constrained by current hardware limitations.

2.3 Auditory Processing Streams: Where vs What

Designing interactions for users to select sound elements requires an understanding of how human auditory perception works. The *Dual-Processing Streams* theory is the mainstream understanding of how humans process auditory information, with the processing divided into "where" and "what" streams [74]. While these two streams work in parallel and are ultimately integrated into a single auditory object [5, 25], they represent distinct mental modes of how people understand and navigate their sonic environment.

The "where" pathway, or the dorsal stream, is responsible for localizing sound sources and understanding movements of sounds in 3D space [74, 75]. This spatial pathway reacts rapidly within 10–30 ms [2], making spatial identification feel instinctive and low-effort. However, the dorsal stream is highly susceptible to spatial ambiguity, as there is an inherent limitations for human's auditory spatial localization. One of such is the "cone of confusion", which is an area extending outward from the ears that makes it difficult to identify the front/back and left/right location of the sound sources [11, 18, 53]. Vertical localization is another limitation, as differentiating the elevation of sound sources relies on the subtle frequency differences shaped by the outer ear structure [11, 18, 53].

The "what" pathway, or the ventral stream, is responsible for the linkage of sounds to humans' semantic understanding [24, 74, 94]. Gaver [22] suggests that humans naturally group everyday sounds into semantic categories based on our experiences with the sounds in our everyday life. These semantics of sound forms by how we associate different acoustic properties (e.g. pitch, timbre) to the object, material,

action, and environment we observe when the sound occurs [8, 23, 47]. However, semantic categories have a certain level of abstraction, which, in reality, there can be multiple sound sources under the same category, making it hard to have a fine-grained differentiation [23, 26]. Additionally, categorization of sounds is a subjective process, and thus people's semantic groupings can differ from each other [46, 94].

These two distinct processing pathways suggest two approaches for designing selective noise control interfaces. A spatial interaction principle based on the "where" stream allows users to select and adjust sound sources by their location, requiring a spatial and location-based interface design. A semantic interaction principle based on the "what" stream allows selection through predefined categories that encapsulate users' associations with sounds, leading to a category-based interface design with textual semantic information. These two principles form the basis of the two interaction principles compared in this study.

2.4 Visual Interface for AAR Auditory Manipulation

To facilitate interaction for AAR selective audio manipulation, including selective noise adjustment, it require modalities outside of hearing for its control. One of such modalities is vision, where the idea of using a visual interface for AAR has been envisioned back in 2003 and has since been researched and realized [35].

Haas et al.'s [28] study that compares screen-based and head-based interaction for selective sound adjustment have used a list of semantic categories of the sounds for their screen-based interaction, which the result shows it has higher usability and lower cognitive load in all conditions compare to head based interaction. Dynamic (i.e. sound option appears when it is detected) and static (i.e. a predefined list of sound options is provided) semantic interfaces are tested with their phone-based Aurchestra system, where users have quicker response time for the dynamic interface but do not allow adjustments in advance that some users preferred [65]. Veluri et al. [90] have developed another phone-based system called Semantic Hearing and compared pre-define toggles and prompt based 2D interface, which both have its limitation, with the toggle-based interface having too much options and prompt based interface have misinterpretations that cause wrong adjustments.

Meanwhile, the ARCube [82] and a patented AR audio mixer [59] system have been designed to allow the placement of audio objects (i.e. spheres) in a 3D space and the access of a physical audio mixer/MIDI controller for more intuitive spatial audio creation. AudioMiXR [93] is another system that allows the same 3D audio object placement, but the reason for AR usage focus on situation awareness. These systems are more focused on manipulating virtual sound sources and blend it with the real-world soundscape. However, recently, Xu et al. [97] have developed the MoXaRt system that uses audio separation and visual detection to allow selective sound source adjustment, which has an interface that displays a visual outline marker to indicate the interactable sources.

These works can be categorized as semantic or spatial visual interfaces based on the two auditory process streams, each of which demonstrates different inherent design properties. Semantic interfaces offer efficient category-level control through familiar 2D GUI structures with toggles, sliders, and buttons, while spatial interfaces leverage 3D space for dynamic and direct object-based source manipulation. However, as these works all focus on a single interface design, how the two principles differ in user experience remains unexplored. This paper addresses this gap by designing and comparing both interfaces under controlled conditions to understand how the two interaction principles differ in user experience.

3 USER STUDY

3.1 Study Design

A within-subjects study design was employed with two conditions of interfaces designed based on the two interaction principles which each encapsulates both the auditory process pathway and the relevant implicit design properties. The *spatial interface* requires participants to select sound sources by directly interacting with source objects, and the *semantic interface* requires participants to select noise by pre-defined categories. To avoid overloading participants with a large amount of sounds, all sounds were introduced in pre-defined sequences of delay time (see Table 1). The types of sounds included are covered and considered as common in most the office sound research [1, 17, 20, 42, 70, 71]. Other than the ventilation and outdoor sound being consistently played from the beginning, other instantaneous and event-based sounds are scheduled randomly, as in real office environment, these sounds also happen in a more random manner. The interface order and sound sequences were counterbalanced across participants to mitigate order and learning effects. The office environment, the number and types of sound sources, and volume adjustment interaction were kept identical across both conditions. The interface study was conducted before the post-study interview, as it is necessary for participants to experience such a selective noise control system to meaningfully reflect on its real-world applicability. A pilot study with 3 participants was conducted prior to the main study to refine and validate the interface design and study procedure.

3.2 Participants

20 participants with an average age of 27.1 (SD = 3.91, min = 23, max = 37) were recruited for the study. None of the participants reported having diagnosed hearing impairments, and all reported normal or corrected-to-normal vision. Participants were recruited through convenience sampling via personal networks and group chats. Pastries were offered as compensation for participation. All participants had experiences of working in open-plan office environments, with 5 currently working in an open-plan office and 15 had worked in one within the past one to five years. One participant reported having no experience with VR/AR, while all other participants have experience with VR/AR, ranging from occasional to frequent usage for entertainment, development, and research. One participant reported that they did not wear any headphones in their workplace. All other participants uses headphones for noise coping, with 13 played masking sounds (e.g. music, white noise), 5 alternated between masking sounds or only ANC, and 1 used only ANC.

3.3 Prototype Design

Participants tested the two selective noise control interfaces in a VR mockup open-plan office scene (see Figure 1) with a simulated auditory environment. Simulations of AR interfaces in VR has been used to ensure consistency and bypass technical limitation, which allows researchers to focus on evaluating the interfaces rather than the technical implementation [29, 45, 72, 85, 92]. Similar with auditory environment, simulated soundscapes have been used in noise related research and used with VR environment to test auditory spatial awareness [19, 80, 100]. Although VR simulations are used for AR research, there is not yet a standard for how the VR environment should be designed, including for office environments. A relatively small and less complex office scene is choose here to also avoid overload.

The two interfaces follow the inherent design properties of spatial and semantic interaction principles. The spatial interface requires participants to directly point and select sound source objects, where they

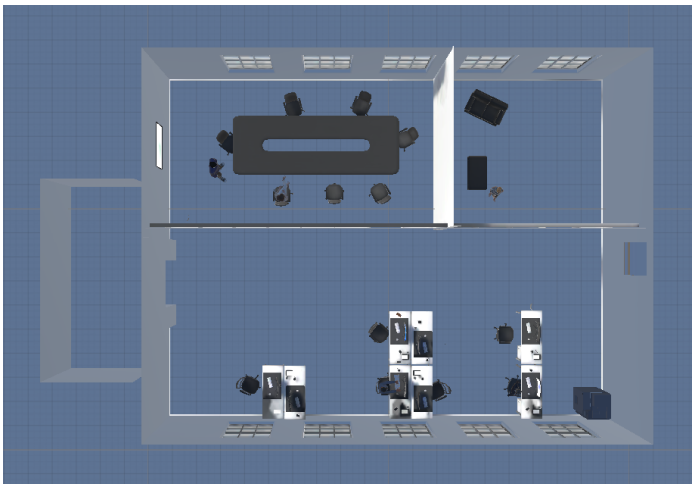


Figure 1: VR open-plan office scene and sound source placement.

are dynamically selectable only when sound is emitting. Similar to the MoXaRt system [97], outlines are used to indicate active sources, and a volume slider is shown next to the source object (see Figure 2). The semantic interface has a 2D panel with buttons of predefined noise categories that is anchored on the table (see Figure 3), and participants need to select the category button the sound belongs to. The categories of noises are synthesized from existing office noise research.

To avoid visual overload, the volume slider only appears when a source object or category button is selected. As precise adjustment is not the focus of this research, the volume has discrete level adjustment (0, 25%, 75%, 100%) rather than continuous, which do not have noticeable usability differences [28]. Controller-based interaction is used to reduce hand tracking errors and lower the learning curve for participants with limited VR experience. The interaction for adjusting the volume requires participants to hold the trigger button after selection and move the controller up and down, and this volume adjustment interaction is consistent for both interfaces.

3.4 Apparatus

The study was conducted in a quiet room with sound-absorbing walls to minimize external noise. Participants are seated on an office chair where they can rotate and move around in the VR mock-up office environment. Meta Quest 2 headset and the Bose Quiet Comfort Headphone were used for testing the two interfaces. The prototype is developed with Unity 6000.0.65f1 with pre-made 3D model assets for the office environment and sound source objects. Unity's Audio Spatializer SDK is used for Head-Related Transfer Function (HRTF) audio spatialization [89]. Ten office sound source objects are placed based on a real-world office layout (e.g. typing from a neighboring desk, ventilation near the ceiling). The audio clips for the sound sources are downloaded from Freesound.org under Creative Commons 0 or Attribution 4.0 licenses (attribution in Appendix D). The volume of the audio clips is normalized through Audacity's loudness normalization tool and adjusted according to the decibel difference of the noises based on office noise research (see Table 1) [17, 20, 42, 70].



Figure 2: Spatial interface when Keyboard is selected.



Figure 3: Semantic interface when Electronic category is selected.

Category	**Sound**	**Decibel** (dB)	**Sequence 1** (s)	**Sequence 2** (s)
Building	Ventilation	30	0	0
	Outdoor	20	0	0
Speech	2 people conversation	40	30	15
	1 person phone call	40	105	45
Electronic	Keyboard	35	15	30
	Printer	30	60	120
Notification	Phone ring	40	45	90
	Video call software ring	40	75	60
Movement	Footsteps	25	90	105
	Rolling chair	25	120	75

Table 1: The office sounds used in the study, their associated category, and delay time.

3.5 Procedure

For each participants, they first got a briefing about the study, asked to sign a consent form, and answered background questions about their age, experience with open-plan office, and VR/AR familiarity. Before

experiencing the two interfaces, participants had time to familiarize themselves with the mock-up office scene and let the researcher know when they were done.

For each of the two conditions, the researcher first briefly explained how the interface works for selecting and adjusting the sounds. Then the participants entered a training scene that had 3 cubes and 3 spheres as sound sources, where 3 of the 6 sources were randomly picked to emit the sound of "this is a cube/sphere" and participants could adjust the volume of these abstract sound sources. Participants are asked to inform the researcher when they were familiar with the interface, and the researcher then loaded the office environment. Participants had 3 minutes to use the corresponding interface to adjust the volumes of the different office sounds to their liking. During this period, they were encouraged to think aloud and interact with all sounds they heard to help the researcher observe interaction behaviors. After the adjustment experience ended, they filled in two questionnaires. One is the NASA TLX questionnaire [32] (see Appendix A) for cognitive load. The other one is a 7-point Likert scale custom questionnaire (see Appendix B) that evaluates participants' perceived usability (understandability, ease of use, intuitiveness, expectancy) and performance (sense of control, identification accuracy, effectiveness, and efficiency) with the selective noise control interfaces [6, 14, 63, 64].

After participants finished both conditions, a post-study semi-structured interview was conducted. The first part of the interview focused on understanding participants' likes, dislikes, and overall preference of the two interface and the reasoning behind them. The second part focused on the real world application, where they were first asked about existing experience with the office noise environment and their ways to cope with it, and then about how they perceive themselves using such a selective noise control system in their real workplace and compare it with they existing coping mechanism.

Each condition took roughly 10 to 15 minutes, including familiarization, exploration, and questionnaires. The post-study interview lasted approximately 30 minutes. After the interview, a debriefing is done to let participants know what the study is about. The entire study took around 60 to 75 minutes.

4 RESULTS

In order to compare the user experience of the spatial and semantic interaction principle for selective noise control, data on overall satisfaction, cognitive load, usability, and perceived performance were collected via preference counts, NASA-TLX, and the custom interaction questionnaire. Linear Mixed-Effects Models (LME) are used to analysis overall scoring of the questionnaires. Median and interquartile range (IQR) were used to analyze each NASA TlX subscale, while counts of different Likert score rating were used to analyze each interview questionnaire questions. A significance threshold of $p \leq 0.05$ was applied for all LME analyses.

Interview data for each interaction principle were analyzed separately with hybrid thematic analysis [21]. Initial deductive codes were derived from the NASA-TLX and custom interaction questionnaire dimensions (see Appendix C). The majority of the final codes were inductively developed from the data, either extended from the deductive codes or emerged independently. Each interaction principle produced sub-themes that falls under themes related to sound source identification method (where and what) and interface design characteristics, while additional themes emerged outside of these two areas. Inductive thematic analysis was used to analyze the real-world perception interview data, following Braun and Clarke's [9] six-phase approach.

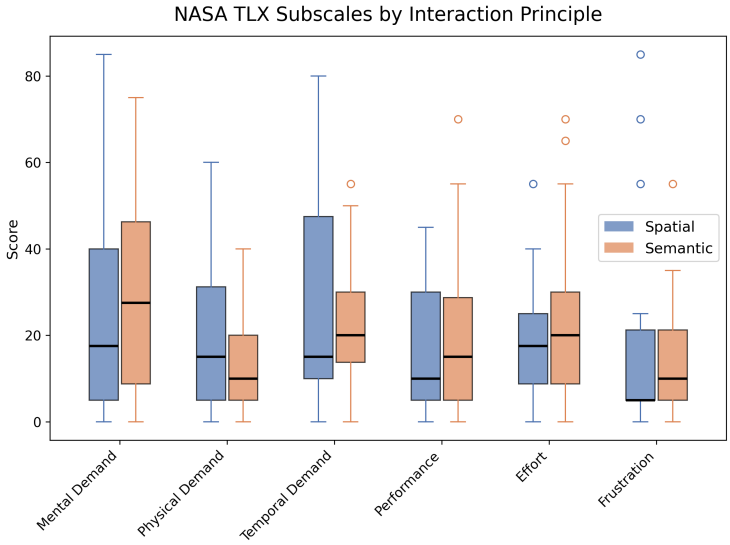


Figure 4: NASA TLX Subscale Score Distribution by Interaction Principle

4.1 Quantitative Finding

The overall preference showed most participants prefer the spatial interface when considering the interaction alone (spatial = 12, semantic = 5, no preference = 3). However, when real world work environment is considered, most participants do not have a clear preference, and, for those who have, more prefer the semantic interface (spatial = 2, semantic = 8, no preference = 10).

LME result from the NASA-TLX data showed that there are no significant differences in the overall cognitive load across the two interfaces ($\beta = 0.750, p = 0.818, 95\%\text{CI}[-5.646, 7.146]$). As for each cognitive load subscale, the spatial interface had a lower median score on mental demand, temporal demand, performance, effort, and frustration, but higher on physical demand compared to the semantic interface (see Figure 4).

For the custom questionnaire, LME models were fitted for the overall interaction score, as well as separately for each subscale group for perceived usability (Q1-Q4) and performance (Q5-Q8). The spatial interface had a higher overall interaction score ($\beta = 0.436, p = 0.047, 95\%\text{CI}[0.007, 0.865]$) and usability score ($\beta = 0.463, p = 0.001, 95\%\text{CI}[0.191, 0.734]$) over the semantic interface. However, there was no significant difference in the perceived performance for the two interfaces ($\beta = 0.300, p = 0.294, 95\%\text{CI}[-0.260, 0.860]$). As for each interaction question, the spatial interface showed a higher concentration of higher scoring (i.e. score 6 and 7) across all the question, except for Q2 (Ease of Use) and Q7 (Effectiveness) (see Figure 5).

4.2 Qualitative Findings

4.2.1 Spatial Interaction Principle Themes.

4.2.1.1 Spatial Source Searching. The following themes examines how the mental and physical spatial searching process of sound sources shaped participants' identification process.

Cognitive Locating Process. Although identifying sounds based on location feels intuitive, the identification process is not that easy and requires trial and error (P12: "I think there is definitely a challenge

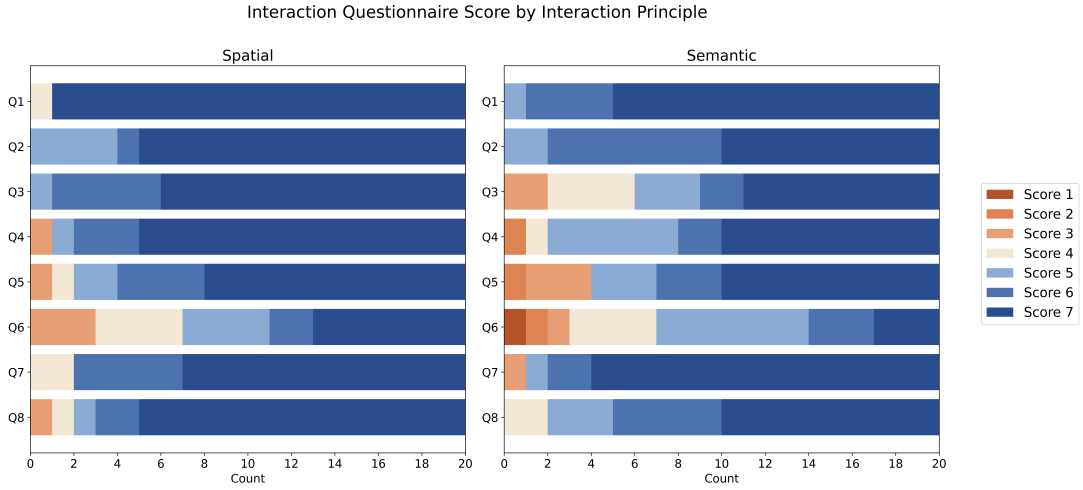


Figure 5: Interaction Questionnaire Score Distribution by Interaction Principle

in locating the noise, so that requires a lot of trial and error"). Participants also noted there is a considerable effort needed for locating, as one described, "There would be some amount of time spent for me to try to locate every sound and adjusting that to my liking" (P19). For sounds that were difficult to locate, such as the phone ring and the ventilation, this effort could increase mental load and even escalated to frustration (P20 : "When I cannot find which phone is making the sound, it just drive me crazy ... when I cannot identify where the sounds come from, it feel like mental load is just go up a lot").

Visual Navigation of Sound. Participants noted that it is a natural reaction for them to look at the source when they hear a sound (P10: "It's natural for you to go [where the sound is emmiting from] physically and have a look"). Seeing the source also gave them a visual confirmation of what they were hearing, allowing them to be more confident that they were adjusting the right sound, as one noted, "I was looking at the object, like the phone that is making a sound, and I know was reducing it. So I was like fully confident" (P04). However, this visual navigation came at a physical cost, as it requires participants to physically turn around to locate the source (P06: "I might not want to spend time turning my chair around to find the sound").

4.2.1.2 Object-Based Interaction. Participants commented on how the characteristics of object-based shaped their sound sources selection interaction experience.

Direct Selection. Pointing at the source directly feels intuitive to the participants, with one comparing it to a remote controller: "It feels like turning down the volume of a TV by pointing the remote to somewhere" (P05). Also, being able to adjust the sound source directly also gave them a good sense of control (P17: "[The spatial interface] makes you feel in control because you are directly adjusting the sound from themselves"). However, one participant had an opposite experience, as they noted: "If I have to click on and adjust the noise [directly], that means they (the noises) could control me rather than [I] control the noises" (P18). Participants also mentioned that there were difficulties on selecting moving or occluded sources, as one noted that "if a guy on another floor and his phone was always ringing, so we could hear it down there. I would not be able to do mute his phone with the direct pointing" (P05).

Individual Adjustment. Interacting with each object individually makes participants feel both intuitive and in control, with one participant noting, "I can sort of choose [the sounds] very specifically to get

that noise out of the way. That made it feel very much like I am in control of the situation" (P19). This also offered participants high control granularity (P03: "When there are multiple sources of the same sound, it is better as I can adjust one instead of all"). However, this one-to-one interaction became a burden in dense sound environments, where some participants felt overwhelmed (P01: "I remember being overloaded with the interactions, like, oh here is this and here is that, I have to turn this and this down").

Social and Practical Concern. Participants have expressed the discomfort and social awkwardness of using interactions that target a person, as one participant noted, "If you were to point at them or some other interaction where you just look at them, then that also becomes a little bit awkward. It is not very normal to look at somebody and then be like, oh, I am looking at you because I do not want to hear you" (P06). Having a coworker relationship with the person you are pointing to worsens this social awkwardness (P05: "I don't care about pointing at the printer. I care about pointing at Susan who sits next to me every day"). Another concern participants have is the need for AR glasses, as they are not used to wear any eye-wears (P01: "For the [AR] headset part, I think if I wear it for too long, I will feel uncomfortable, since I don't wear glasses").

4.2.1.3 Environmental Experience. This theme captured how both the searching and selection process shaped participants' experience in the environment. Participants described a greater sense of connection with the environment during the process of scanning and locating sources (P19: "You are scanning around, you are looking for the things that cause the sounds ... I feel like I am more present in the environment"). This process also became an entertaining experience, with one participant noting, "It became a little bit like a game to understand where this sound is coming from and how should I get rid of it?" (P16).

4.2.2 Semantic Interaction Principle Themes.

4.2.2.1 Semantic Understanding and Categorization. Two themes have emerged from how participants' sound selection process is defined by the semantic category-based structure.

Category Ambiguity. The semantic categories were considered ambiguous and mentally demanding to participants, particularly for the keyboard, ventilation, and phone ring sound (P10: "I was a bit confused by the sound of the typing. I thought it was in movement, because it kind of relates to human movement, but apparently it was in electronics"). This ambiguity led to some participants needing to guess which category a sound belonged to, with one noting that "Every time when it (the adjustment) worked, I felt like I got lucky" (P04). This resulted in trial and error, requiring more time and causing frustration, as one participant noted, "Sometimes I need to raise the other noises to check [if the right category is selected] ... it gets more annoying and a little bit more frustrating, because I need to hear the again and it makes myself more disturbed" (P14). Participants also noted that the categories did not align with how they understand different sounds (P06: "I do not always consider it (the sound) as the same category as somebody else").

Selection Granularity. Participants noted that the interface offered less granular control as each category button covered multiple sounds, as one participant noted, "I did not have as much of a granular control ... I feel like the phone conversation is super distracting to me. When there's a discussion, that's sort of something that I can live with. But I will mute all just because I am annoyed by one of them" (P19). Yet, this same grouping of sounds allowed participants to adjust multiple sources quickly and with a single interaction (P03: "I think for the noises that I do not want to have at all, like the very loud noises,

I can shut down all at once"), and reduced the mental load of identifying sounds individually (P15: "It is nice when they (the sounds) are grouped because then you do not have to think about them one by one"). To balance the low granularity and group adjustments, a few participants suggested having both broad and specific categories for the interface (P11: "I want to be able to lower it (the sounds) broadly. But maybe there could be a toggle to open up a menu, and I can lower just this or just that").

4.2.2.2 2D Interface. The following themes relate to how the design of the fixed panel interface influenced participants' interaction experience.

GUI Structure. Participants noted that listing out all sound categories allows them to have an overview of what sounds are expected to be present in the environment, giving them a sense of control, as one participant noted that "All types of noise are laid out there, I feel like it gives me some more time to know what I want what I want to mute" (P13). Participants also commented on the number of buttons, where the current small amount of buttons layout makes the interface easy to use (P01: "I could easily select, because there were not many categories"), but there can be risk of having too much category buttons that can lead to cognitive overload (P01: "If there would be more categories, then it would be too many to choose from"). One participant commented that there was no clear reasoning on the ordering and structure of the category buttons (P15: "It did not seem there was any logic in the order or position [of the buttons] that were meaningful").

Fixed Location. As the interface is a panel fixed at one location, participants commented that this allows a centralized interface for all sounds, making it easier to control (P16: "All of the audio control is centralized in a specific place. You would be easier control all of it") and required minimal physical movement (P01: "I just need look down when I hear the noise again"). However, this also meant the participants are mostly just focusing on the panel, which reduces their awareness of the environment (P16: "There was a lot of noise that I did not necessarily notice when I was using it").

4.2.2.3 High Efficiency. Many comments from the participants were about how efficient this interface allows them to adjust the noises. Other than the before mentioned quick adjustment for multiple sources, participants noted that this interface also allows them to preadjust noises that they already know that they do not want (P20: "Here I can have a choice at the beginning, like, which noise annoying me the most? I can choose that, like the notifications, so I will can already mute them"). The interface was also convenient after understanding which sounds are in which category, with one participant noting: "I have to try a few times, then I kind of understand [the sound categorization], then it becomes more convenient for me" (P07). However, this efficiency requires a trade-off from the level of control (P03: "Although it is efficient, but then you lose a bit of control").

4.2.3 Real-World Workplace Perception Themes.

4.2.3.1 Soundscape Curation. Participants felt that selectively adjusting sound can allow them to remove annoying noises while preserving sounds that they like or deemed important, such as ambient sounds (P10: "I do not want to mute everything. I really like the building sound, like the sound of the the ventilation") and emergency related sounds (P16: "If the fire alarm starts ringing, I can still hear it and run away from the building"). This could also give them the agency to customize their soundscape that ANC headphones cannot offer, as one participant noted, "My preferences for sound are very contextual, so being able to adjust them specifically would be great... I have more specific options rather than just having no headphones, headphones with music, headphones with noise canceling" (P05). Some participants

also stated that a curated soundscape can replace masking sounds, as one participant mentioned, ”I can decrease [noises] to an acceptable level and still have some stimulation from the environment, so I will not get bored that easily and will not need the extra music” (P14).

4.2.3.2 Positive Social Effects. Compared to using ANC headphones, participants stated that such a system can improve their social interaction with their coworkers. Muting other sounds while preserving speech that are relevant or interesting to them allows them to be aware and easier to join those conversations (P11: ”There is sometimes you want to hear that someone is talking about something so you can join the conversation”). Participants also pointed out that keeping human speech and movement sounds audible allows others to get their attention more easily, as one participant noted, ”When I use headphones, I cannot hear if someone is coming or talking to me. It happens a lot of times that I have a person next to me that wanted to ask me something, but they need to touch me because I could not hear [them]. So for this kind of situation, it would be really helpful” (P02). Some participants also suggested that this noise adjustment can become a conversation starter about their sound adjustments with their coworkers and possibly create new social connections (P14: ”With this kind of system, we can be more aware of the sound around us. It might be a cool topic to share with my colleague, and we can also discuss our configuration”).

4.2.3.3 Implicit Effects. There are some effect participants mentioned that were not directly caused by the noise adjustment itself. Participants stated that being aware of the office sounds and hearing some of them can make them feel more present, as one mentioned, ”[With this system] I could feel that I am still connected to the environment... I think the feeling of presence could help me to focus more and to collaborate with other people better” (P18). Some also think such a system can make people more aware of the noises they make, making people more mindful that they might be causing annoyance (P05: ”I feel like if everyone had it (selective noise control system) maybe people would slowly get quieter”).

4.2.3.4 Removal of Human Speech. Although keeping human-related sounds on can be beneficial, participants stated that one of the major annoyances was human speech and would want to mute them (P06: ”Sometimes you hear stuff that you do not want to hear, like things that made me uncomfortable. It will be nice to be able to not hear those conversations”). However, this can have the same social isolation issue headphones have, as one participant pointed out, ”There is definitely the risk that you shut yourself off from the office [social] environment” (P12). Participants commented that wearing hearables while still being able to hear others breaks the social cue of ”wearing headphones equals cannot hear them”, as one participant mentioned ”[Wearing headphones] gives a pretty clear signal of I am not to be disturbed by anything right now ... I guess for a selective one, you can be already hearing something, but then people still see that you are wearing the headphone” (P11). There is also the concern that muting an individual coworker could worsen their relationship, as they can feel singled out, as one mentioned: ”When you do it (sound adjusting) to inanimate objects, it is okay, but if you do the humans, how do you not take this personally?” (P04).

4.2.3.5 Adjustment Increase Workload. As users needed to adjust the sounds manually, some participants stated that this became an additional task (P14: ”I need to adjust everything on my own, so it becomes more work”). Participants also felt the need to make adjustments frequently when new sounds appear, which can cause distraction from focusing on their work, as one participant commented, ”I do

not really know how that fits into a workflow for me, whether I want to consistently break out of what I am working on and start trying to identify what is making a noise" (P05).

4.2.3.6 Possible Issues. Participants raised several concerns about using such a system. They worried that the aforementioned issue with muting specific coworkers can lead to bigger workplace harassment issues, such as workplace bullying and discrimination, as one mentioned, "I feel like it is kind of bullying when I would be able to block out someone that I do not like. Let's say if I am racist or if I am xenophobic, maybe I am leaning more to blocking out these people, then that can be a problem in the workplace" (P16). Some also worried that, if such a system is normalized, the systematic issues of workplace noise might become individual responsibility (P05: "I feel like maybe it takes a bad working environment and puts the responsibility of mitigating that to workers instead of putting up separation walls, making meeting rooms soundproof, teaching people to be respectful to each other"). Long-term effect on their noise tolerance and hearing is another concern, as one noted "I would become much less resistant to certain noises and I would get easily frustrated by them when I hear it in other places" (P09). There is also some privacy concerns due to how such system will need constantly listen to sounds in their work environment, violating individual privacy (P04: "I feel like you have a right to know that your sound is being listen when there is monitoring happening at all times") and company information security (P17: "This system will detect the noise by its own. What if the office is like working for some secret, then the data might be stored or leaked").

4.2.3.7 Design Desire. Several features were suggested by participants as desired for selective noise control systems. As participants desire to minimize the time and effort needed to adjust their soundscape, they stated that they want the system to allow adjustments in advance (P14: "If you can adjust in the beginning, like pre-set everything and already customized, then it will be pretty good") and saving their adjustment setting of different types of sounds (P05: "Maybe you could have favorites, like saved settings for when you hear keyboard then lower it"). Participants have also suggested functionalities that extend from just reducing and muting selective sound, such as selective amplification (P03: "It would be nice if you can tone up the volume for specific things") or muting as the default (P08: "It can mute everything first and have option to hear certain thing ... So it could be like mute all and then select to hear one person speaking"). Combining with playing music is another functionality participants desired, as music is not only a masking tool but a focus cue (P03: "For me it (listening to music) motivates me and sets me into like a mindset of working ... I would combine them (selective sound adjustment) with music"). Some participants also bring up the possibility to include adaptive features, specifically automatic unmute when people are approaching them (P06: "I had the issue where people tap me. The other option here is maybe that this kind of system where the system turn the sound on by itself if somebody want to contact me").

5 DISCUSSION

5.1 Interaction Principle Comparison (RQ1)

The findings for RQ1 suggest that the two principles offer distinct experiential qualities examined across three aspects: interaction workload, perceived usability and performance, and overall user preference.

Although no significant difference in overall workload was found between the two interfaces, the subscale medians and qualitative findings suggest differences in specific dimensions for each principle. The spatial interface had a lower mental demand, which aligns with how the "where" auditory process stream

has a quick response time which requires less mental effort for sound source identification compare to the ”what” pathway [2]. However, the spatial uncertainty of the sound localization process remained as a source of cognitive demand. The ventilation, located near the ceiling, was one of the sources that were more difficult to locate, which matches with the vertical localization limitation [11, 18, 53]. Yet, the ”cone of confusion” effect [11, 18, 53] did not appear to affect the identification process, possibly because the visual confirmation of sources within the participants' eyesight compensated for front/back and left/right localization ambiguity. This visual confirmation is necessary and becomes an issue when the source is not clearly visible, such as smaller or distant sources like the phone, which participants had trouble selecting and adjusting. Visually occluded sources, such as sources behind walls, are also not selectable with the spatial interface [28]. Other than the visual needs for selection, participants also brought up the system's limitations on targeting moving sound sources. The need to physically orient toward the source for the selection explains why the physical demand is the only NASA-TLX dimension that is higher for the spatial interface. The high overall workload for the semantic interface is likely due to participants' semantic association with sounds differed from the ones offered in the interface [46, 94]. This required participants to go through a translation process and trial and error to verify whether they have adjusted the right sound, which also explains the higher mental demand and frustration scores.

The spatial interface scored significantly higher on the overall interaction score, likely driven by the higher perceived usability score. Both the direct and individual adjustment factor of the objects-based interaction were perceived as natural ways to select sources, consistent with participants finding the spatial interface was easier to understand and use. This also explains the higher expectancy score, as participants connected the sound and the source being adjust straightforwardly, while the semantic interface offers less clarity on which sound is being adjusted. The social and practical issues with the pointing gesture and AR glasses mentioned when participant imagine using such a system in their workplace did not appear to affect participants' perceived usability, which is likely due to the interaction testing being short and done in a virtual environment with no real people. Although participants noted that having an overview of the possible sounds make them feel in control of the soundscape with the semantic interface, this did not appear to overcome the limited granularity, reflected from the lower score in sense of control. There were no differences in selection accuracy and effectiveness scores, which might reflect that both interfaces required some trial and error for each of their identification and selection limitations. Even though participants have commented that the semantic interface offered high efficiency, the efficiency score is lower than that of the spatial interface, which might due the learning curve participants described, where familiarity with the category structure was needed before the interface felt efficient.

The overall interaction preference, where the majority of participants preferred the spatial interface, is consistent with the lower workload and higher usability findings discussed above. The sense of environmental connection and enjoyment might also contribute to this higher preference count. This higher preference may also be partially contributed by novelty effect [52, 58], as the spatial AR interface was a less familiar interaction paradigm for most participants compare to a traditional 2D interface. However, when the real-world work context was considered, this preference shifted toward the semantic interface. This suggests that interaction quality alone does not determine adoption preference in workplace settings. Factors such as efficiency, social acceptability, and integration into existing workflows become prioritized over intuitiveness and sense of control when users consider long-term daily use.

5.2 Real-World Perceived Usage (RQ2)

Overall, the findings suggested participants saw potential for such a system to improve their workplace experience when compared to using ANC headphones, along with practical concerns for actual adoption.

Participants agreed that selective noise control could extend workplace well-being beyond what ANC headphones offer. Selective noise control shares the same effect as ANC headphones on removing unwanted noises to improve concentration and reduce stress. However, unlike ANC, which offers binary on and off noise cancellation, selective noise control could give participants agency over their sound environment and curate it according to their own preferences and context. The ability to control and customize one's environment has been known to benefit individual's well-being, as it fulfills the emotional need of control and choice and can further reduce their stress [48, 99]. Other than explicitly being in control, selective noise control also allowed participants to feel more present and connected to their environment by encouraging participants to interact with the office space and maintaining some degree of awareness of their soundscape. This can prevent the sensory detachment and isolation issue ANC headphones can cause [57, 81].

Improvements in social interaction with their coworkers are another potential benefit selective noise adjustment system can offer. It can facilitate both casual and work related communications, such as small talk and coworkers need of their attention, by keeping human-related sounds (i.e. speech and footsteps) audible. This can extend open-plan office design's communicative and collaborative goal beyond visual contact by including omnidirectional auditory awareness while still being able to reduce noise annoyance [20, 36, 40]. However, speech remains one of the more commonly reported annoying sounds participants wanted to remove [1, 20, 70]. If the system removes relevant or all speech, then it also removes the benefit of improving communication, having the same social isolation concern that ANC headphones have. Specifically for allowing others to get their attention easily, participants suggested that the system to automatically detect people approaching and unmute their sounds.

Selective noise control may also negatively affect workplace social relationships. Gesture-based control for the spatial interface was one of the concerns, as participants stated the social awkwardness of using it in the workspace. Although such expressive gestures are considered relatively socially acceptable, as others would know what the users are doing [77], using them in the workspace with coworkers in the environment is considered less acceptable compared to private settings [61]. Furthermore, as adjustments can be directed towards a specific person and participants have an existing personal relationship with that coworker, this amplifies the social discomfort as it may be perceived as being passive-aggressive for being specifically targeted for muting. Beyond direct interaction with coworkers, the system also creates a broader shift in how the sonic environment is shared among coworkers. With such a system, each person can have a different perception of the office soundscape based on their individual adjustments, and it removes the visual social cue that headphones offer [86], leading to others not knowing if an individual is hearing or not. This asymmetric auditory experience creates an unequal information flow and the loss of shared auditory reference that can disrupt existing communicative norms [12, 69].

Participants also raised concerns about the widespread adoption of selective noise control in the workplace. Compared to simply putting on ANC headphones, the need to manually adjust their soundscapes could become an additional daily task for participants beyond their normal work. This may explain why there is a shift in interaction principle preference, as participants expected the semantic interface would allow more efficient adjustment in the long term. Participants also suggested that such systems should allow pre-adjustments and save sound settings to minimize the friction that can add to their workload. More

serious concerns that apply to the broader workplace also emerged. One such concern is potential workplace harassment, extending the targeted adjustment issue mentioned previously. With such a system, people could remove a coworker's voice out of personal bias rather than noise discomfort. This can lead to workplace discrimination and bullying that may be difficult to detect, and thus create or perpetuate a toxic workplace that can have a significant impact to workers' mental health [66, 79]. Another concern is how such a system can shift responsibility for maintaining a quiet workspace away from organizational measures and collective consideration, such as companies installing sound-absorbent materials in the space and coworkers maintaining shared norms of workplace quietness, to individuals. While this issue already applies to ANC headphones, selective noise control makes it even easier to mitigate the office noise issue individually, which could deepen this problem.

Other concerns participants raised extend beyond the workplace context to the general daily use of selective noise control systems. Long-term use was suspected to potentially reduce tolerance for specific sounds, an issue ANC headphones already have [39], but might extend to specific sounds becoming triggers. Selective noise control also raises the inherent privacy concerns of pervasive AR, as constant sound monitoring is required for sound recognition and separation. Privacy and data collection concerns are challenges for AR systems in general [78], and for selective noise control in the workplace, these concerns operate at both personal and organizational levels, including a lack of bystander consent for their sounds to be captured [86] and the risk of sensitive company information being stored and leaked.

5.3 Interface Design Considerations

For the spatial interface, the social discomfort caused by gesture control could be mitigated by adopting secretive gestures [61], such as a combination of gaze and micro-gestures [49, 84]. Although micro-gestures for pointing and selecting object are well researched [41], how to integrate volume adjustment to such micro-gestures still remains an open question.

For the semantic interface, offering both broader and specific category levels can increase the control granularity while preserving the efficiency of controlling multiple sounds at once. However, careful consideration for interface organization is needed to prevent option overload as the number of categories increases, which a hierarchical structure presents as one possible design choice. As for the mental effort needed for sound to semantic translation, allowing users to label sounds based on their own semantic association can be a way to overcome this issue, which the ProtoSound system developed for deaf and hard-of-hearing individuals for customizing sound recognition models offers a relevant reference for this approach [34].

Even though specific design choices for the specific interface can improve their usability, each interaction principle's characteristic introduces an inherent limitation that cannot be resolved by improving each interface individually. Combining the two interaction principles could compensate for these inherent limitations, such as complementing the spatial interface's granular and source-specific control with the semantic interface's efficient category-level adjustment. Such a hybrid system would also allow users to prioritize interaction modes according to their contextual needs, such as using semantic interaction for pre-adjustment of known recurring sounds and spatial interaction for dynamic or unexpected sources. For instance, a hybrid interface could display the list of semantic categories and highlight the spatial location of sound sources belonging to the selected category, which can utilize both the semantic interface's efficiency and the spatial interface's visual comformation. However, how to effectively integrate the two principles in a single interface still remains as an open design challenge of further investigation.

5.4 System-Level Design

Participants expressed the desire to minimize the daily friction of manual soundscape adjustment while retaining a sense of control over their sound environment. This reflects a broader tension between manual and adaptive soundscape augmentation identified in previous work, where users value both the agency of manual control and the convenience of adaptive automation [86]. A middle ground approach of allowing users to pre-configure sound preferences and the system adapts within those defined limits while still allowing manual adjustments could address both needs. In a workplace context, this could take the form of allowing specific adaptive features, such as automatic detection of people approaching or seeking attention participants mentioned, to preserve important social awareness without requiring constant manual intervention.

Selective amplification and music integration are features participants desire beyond volume reduction. Work on broader everyday noise and soundscape personalization has also explored such features [10, 86]. In a workplace setting, these features serve specific purposes beyond general personalization, where amplification can enhance the audibility of the person they are talking to, and music serves as a focus cue rather than purely for enjoyment. Beyond amplification and music integration, a mute-all-as-default approach with selective hearing presents an alternative system design that is useful in the workplace, particularly when most sounds in the office are perceived as unwanted noise, allowing more practical and focused control on specific important sounds, such as relevant speech or emergency sounds, with minimal effort.

5.5 Social and Ethical Considerations

While adopting less expressive gestures can reduce social discomfort, the workplace context introduces social challenges that extend beyond gesture design. Unlike public or private settings, the workplace lies in the intermediate area. Hall's proxemics theory positions the workplace within the social space zone, where gestures and eye contact are easily noticed by others and individuals feel obligated to that person's presence [31]. Other than proxemics, the relationship with coworkers is not as familiar as family members or friends, which gives individuals higher confidence in using gestures, yet not as detached as interactions with strangers [77]. Therefore, in a workplace, even subtle sound adjustment gestures directed at a coworker, such as looking at them, can be noticeable and carry social weight regardless of how discreet the gesture is. Although design can minimize the visibility of spatial interaction based noise control, it cannot fully eliminate its impact on workplace social dynamics. Long-term adoption of such a system would require a new norm to be formed, where the act of adjusting a coworker's sound is understood as a practical tool rather than a personal or social signal.

The asymmetric auditory perception is another factor that influences the broader workplace social dynamic. This relates to the broader perception fragmentation and information mismatch issues AR systems introduce in general [12, 69]. Such asymmetric experience disrupts existing communication norms in ways that are invisible, as there are no visual cues to indicate someone is partially listening to the environment and what they are listening to. Designing a system that makes one's auditory configuration visible to others is a possible solution, but this brings up privacy issue as users might not wish to disclose which sound or coworker they have chosen to mute. Another possible way to mitigate with design is to include a visual affordance on hardware, either on the headphones or AR glasses, to signal that selective hearing is active. However, this only signals that selective hearing is active without revealing the specific sound they are hearing, leaving some degree of asymmetry to remain. Ultimately, if selective noise

control is to be widely adopted, asymmetric auditory perception may need to be accepted as a new characteristic of shared sonic spaces, requiring also social norms to adapt accordingly.

The ability to target and mute individual coworkers that could perpetuate existing workplace harassment and discrimination, the shift of noise management responsibility to individual workers, and the individual and company privacy and consent for pervasive listening are all social and ethical concerns that would arise from deploying selective noise control systems in the workplace. Although selective noise control is intentionally for practical use, like all tools, there is social and ethic implications to is that should not be ignored. These implications cannot be addressed through design or forming new social norms, and instead would require organizational policies and deeper individual ethical considerations on the use of such a system.

6 LIMITATION

The study sample presents several limitations. Participants were mainly recruited from the researcher's personal network, resulting in a relatively narrow age range and a homogeneous background in HCI or engineering. Additionally, a number of participants had limited or no prior experience with VR/AR, which may have contributed to a novelty effect for the preference and higher usability scoring of the spatial interface.

The study design also introduced some limitations. Using a VR simulated office environment imposed limitations on ecological validity, causing some participants to have difficulties imagining real-world usage during the interview. The exposure time for each condition was relatively short (3 minutes), which may have limited the depth of engagement with each interaction principle. Although the sounds selected for the study represent common office sounds, they do not encapsulate all sounds that can occur in a workplace. The office sounds were also spaced out to avoid overwhelming participants, which limited the ability to evaluate how the two interaction principles perform in denser and more realistic noise environments.

Methodological limitations are also worth noting. The think-aloud protocol, while useful for capturing real-time reactions, may have added cognitive load and interaction experience during the tasks and affected participants' experience and ratings. The custom interaction questionnaire was designed specifically for this study and has not been validated against established measures, and its results should therefore be interpreted with caution. Finally, a ceiling effect was observed in the questionnaire 7-point Likert scale scoring, where scores clustered at the upper end of the scale, limiting the sensitivity of the questionnaire.

7 FUTURE WORKS

Further design exploration is needed across multiple aspects of both interfaces, including secretive interaction design for spatial interaction, 2D interface structure, and custom labeling for semantic category, and how to integrate the two interaction principles into a cohesive hybrid system. The manual and adaptive balance design question identified in this study is also worth further investigation, particularly how user-configured adaptive adjustment and automatic detection of attention-seeking interactions can be incorporated. Future studies should broaden the participant demographics and test with more types and instances of office noise. Longitudinal in-the-wild study in a real office setting should be conducted in the future to address the ecological validity limitations and capture system adaptation, changes in interaction preference, and the social dynamics that emerge over prolonged use. Finally, in the future where

selective noise control gets to real-world deployment, how to prevent such systems from unintentionally harming workers, particularly around harassment, noise management responsibility, and privacy, should be looked into.

8 CONCLUSION

Selective noise control systems can be achieved with AR, where interface design can follow the where and what auditory process pathway. Each pathway leads to distinct interface design characteristics, creating two interaction principle: spatial and semantic interaction principles. Although there are already selective sound adjustment systems that uses one of these two interaction principles for their interface design, a comparison between the two principles have not been explored. Such a system also has great potential to mitigate office noises, enabling noise comfort while avoiding the ANC headphones' social isolation issue. However, little is known about how users perceive using selective noise control in a real workplace context, including how it can benefit their work experience, what interface design is needed, and what concerns users have.

The study found that the interface based on spatial interface have higher usability, lower cognitive load, and higher overall preference in isolated interaction, showing that the spatial interaction principle offers a more intuitive approach for selective noise control. However, in real office context, preference shifted towards the semantic interaction principle, due to higher efficiency and social acceptability, highlighting that qualities valued in isolated interaction differ from those prioritized in workplace use. Regardless, each interaction principle has its own advantages and limitations, some of which can be addressed through improved interface design, but neither principle alone is sufficient, which a hybrid approach presents as a possible solution and requires further exploration. Participants also perceived that selective noise control offers a greater sense of social awareness and control over their sonic environment when compared to ANC headphones, indicating its potential to address the social isolation issue. Additional design considerations, system features, and social and ethical concerns emerged when considering deployment in a workplace setting need to be addressed in future work.

As AAR technology continues to mature and open plan offices remain the dominant workplace design, selective noise control represents a promising direction for improving workers' acoustic wellbeing without sacrificing social connection. This paper provides a foundation for understanding how such a system can be designed and how it is perceived specifically in office contexts. The findings and design considerations presented here offer direction for future systems that balance individual noise comfort, social awareness, and ethical deployment in shared work environments, but also broader selective sound control systems.

9 USE OF GENERATIVE AI

Generative AI tools, specifically Claud and Grammarly, were used in this thesis only for grammar and spelling checks and the Swedish translation of the abstract.

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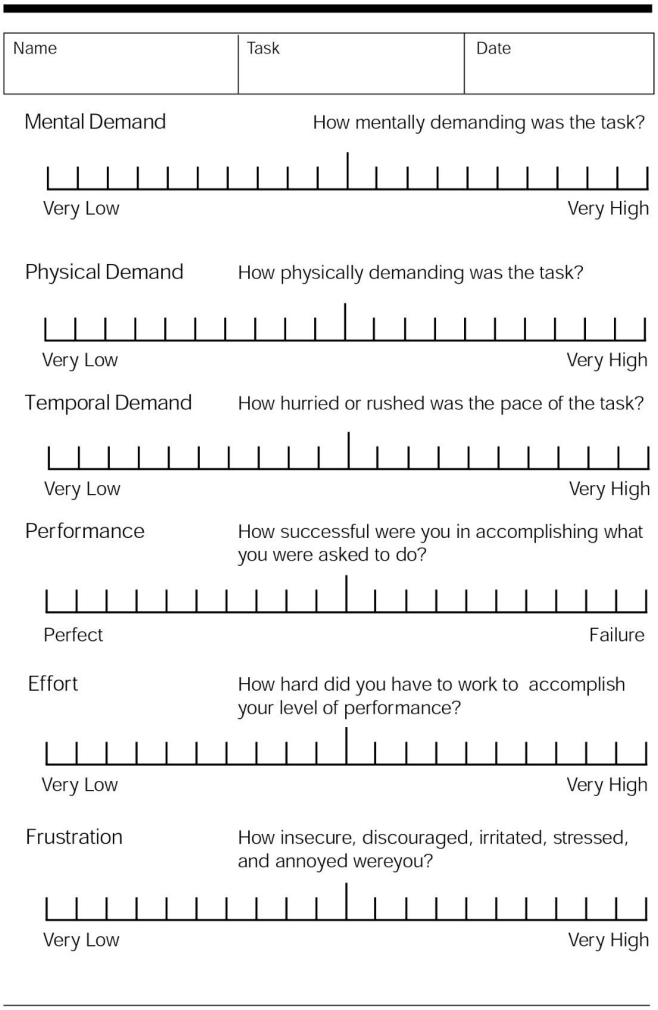
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A NASA TLX SURVEY

Figure 8.6

NASA Task Load Index

Hart and Staveland's NASA Task Load Index (TLX) method assesses work load on five 7-point scales. Increments of high, medium and low estimates for each point result in 21 gradations on the scales.



B INTERACTION SURVEY

Interaction Survey

Name: Task: Date:

		Strongly Disagree						Strongly Agree
Question		**1**	**2**	**3**	**4**	**5**	**6**	**7**
1	I was able to understand how to use the interaction							
2	I found the interaction easy to use							
3	I found the interaction intuitive and natural							
4	The interaction responded the way I expected							
5	I felt in control of the noise around me							
6	I could easily identify the noises I wanted to adjust							
7	I was able to reduce the sounds that bothered me							
8	The effort required to adjust the noises felt reasonable							

C CODE BOOK

Reference	Code	Definition
NASA TLX	Mental Demand	The amount of mental activity required (e.g. thinking, deciding, looking, searching)
	Physical Demand	The amount of physical activity required (e.g. turning, controlling, activating)
	Temporal Demand	The amount of time pressure felt due to the rate or pace of the task
	Performance	The perception of how successful the participant was in achieving the goals of the task
	Effort	How hard the participant had to work (mentally and physically) to achieve their goal
	Frustration	How discouraged, irritated, stressed, and annoyed the participant felt during the task
Custom Questionnare	Understandability (Q1)	How clear the logic and requirements of the interaction are to the participant
	Ease of Use (Q2)	The participant's subjective feeling of simplicity of the interaction
	Intuitiveness (Q3)	The extent to which the interaction feels natural and requires minimal conscious reasoning
	Expectancy (Q4)	The alignment between the user's predicted outcome and the actual system respons
	Sense of Control (Q5)	The perceived sense of agency over sound changes in their environment
	Selection Accuracy (Q6)	The ease of the participant in targeting the specific sound source they want to adjust
	Effectiveness (Q7)	The perceived speed of success of the noise reduction
	Efficiency (Q8)	The perceived effort needed to successfully reduce the noise the participant wanted

D ATTRIBUTIONS

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